

Project Portfolio

Jacky Kam

✉ jkam923@gmail.com

☎ +1 (604)-716-7227

 [LinkedIn](#)

 [GitHub](#)

 [Website](#)

 [YouTube](#)

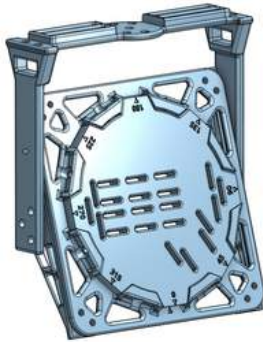
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SEMI-AUTOMATED DATA COLLECTION SYSTEM - XtractOne Technologies

WHAT:

- Provide method for XtractOne to accurately and reproducibly move test objects through security gateways in designated positions and orientations
- Restricted to non-ferrous materials in framing and non-metals for moving components



HOW:

- Designed multi-DoF object holder and structural frame in **OnShape** (later in **SolidWorks**) implementing **DFA** principles
- Spec'd drivetrain components, and coded motor controls using **Python** and **Q-Programmer**
- Fabricated components using **rapid prototyping** methods (3D-printing)

RESULTS:

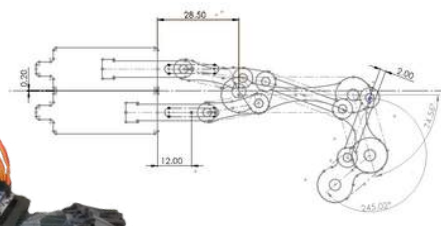
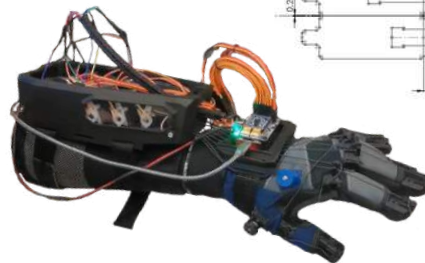
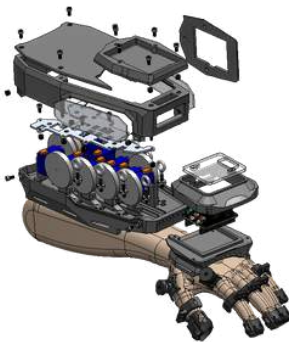
- Mover achieved a consistency of $\pm 0.5\%$ across 20,000 data samples
- Demonstrated stable handling of knives and handguns up to 1kg, at speeds of 1.5m/s
- Independently kickstarted the company's first internal project to be used in the future for QE and installed at manufacturer sites

HAPTIC MANIPULATOR GLOVE - Capstone Project (2024)



WHAT:

- Design a 2-in-1 package haptic controller glove and custom end of arm tool to manipulate objects from a distance
- Accurately track and replicate user motion, be intuitive to operate, and minimize weight



HOW:

- Used **SolidWorks** to create finger linkage calculator, converting linear motion to curling
- Applied **surface modelling** techniques to design ergonomic PCB enclosure.
- Designed tension-based system using **servo motors** to track finger curl.

RESULTS:

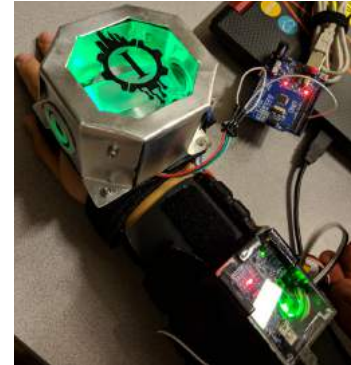
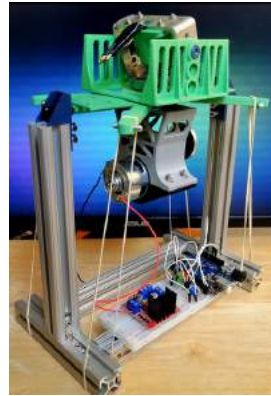
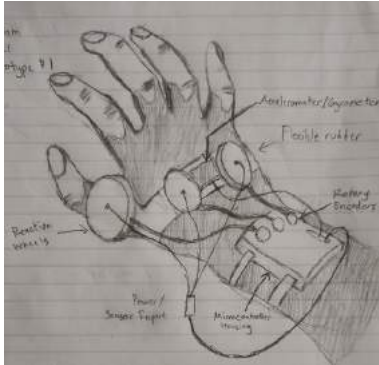
- Tracked 15 DoF of the fingers using under-actuated **four-bar linkages**
- Successfully demonstrated product mimicking user inputs to grasp a variety of everyday objects.
- Placed **1st of 11** teams for innovative design and product showcase.

HAND TREMOR STABILIZER - Capstone Project (2022)



WHAT:

- Devise method to mitigate the outward effects of Essential Tremor in the hands of a teammate
- Actively detect tremor occurrence and differentiate from desired hand movements



HOW:

- Used **Myoware sensors** (EMG) to detect muscular activity and discern tremor occurrence.
- Employed a **testing rig** designed in **SolidWorks** to analyze gyroscopic stabilization effects of a flywheel
- Designed vibration-dampening enclosure, fabricated using **sheet-metal** techniques

RESULTS:

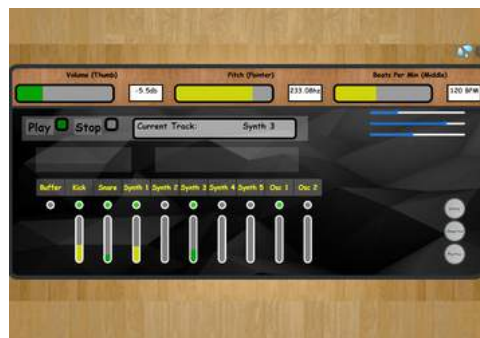
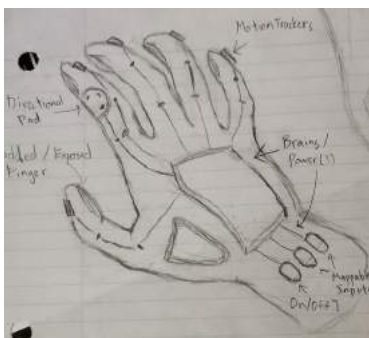
- Project was showcased at the UBC Design and Innovation Day to 300+ participants
- Glove noticeably reduced tremor effects by 60%
- Placed **1st of 13** capstone teams for innovative design and product showcase
- Entered and offered the **Innovation Award** at the Simon Cox Student Design Competition

AUDIO SYNTHESIZER GLOVE - Capstone Project (2021)



WHAT:

- Prototype innovative approach to translate user hand motion into music
- Create user interface to communicate with device and provide tangible visualization



HOW:

- Used **ESP-32** HTTP protocol to send data packets to **web server** over Wi-Fi
- Created user interface in **HTML/CSS** to display audio adjustments and instruments
- Used **JavaScript** to translate data strings into inputs for the UI

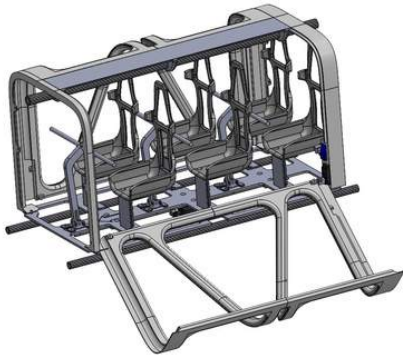
RESULTS:

- Project was able to successfully recognize user gestures and relay information to web server
- Successfully integrated volume, pitch, and tone, as well as various drums and synths
- Placed **1st of 12** teams for innovative design and product showcase

SUBSCALE AIR-TAXI CABIN - UBC Uncrewed Aircraft Systems (2023 / 2024)

WHAT:

- Design realistic scaled-down taxi cabin to be mounted to a fixed-wing VTOL aircraft
- Automate entry, securement, and lighting procedures for 6 passengers (Barbie Dolls)



HOW:

- Guided 8 students through a **carbon-fibre layup** process, including **mold design** and **CFD** analysis before **wind tunnel** validation
- Taught students **surface modelling** techniques for cabin **aerodynamics** and seating design
- Used **SolidWorks** to design compact lapbar restraint using **linear actuator**

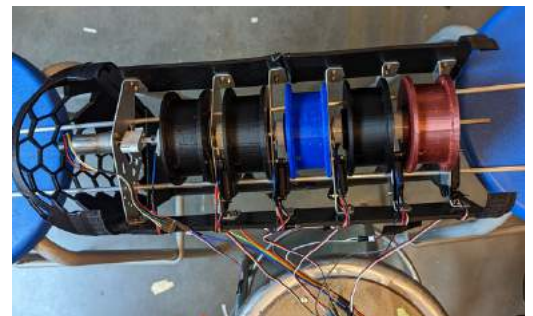
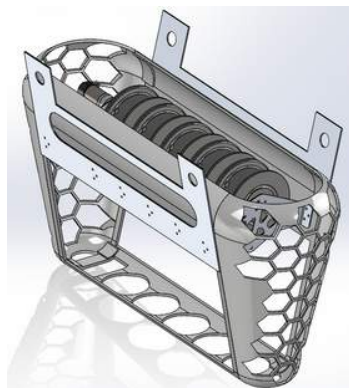
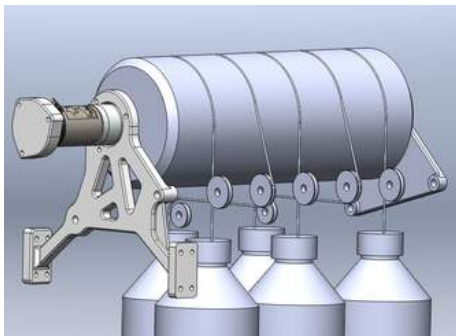
RESULTS:

- Successfully completed UBC UAS' first **carbon-fibre semi-monocoque**
- Use of matrix composites, lightweight PLA, and monokote in construction reduced weight over standard PLA by 70%
- Placed **3rd** at the 2024 Aerial Evolution Association of Canada student competition

UAV WINCH SYSTEM- UBC Uncrewed Aircraft Systems (2023)

WHAT:

- Safely and accurately deploy 5 payloads from 75ft AGL onto 30ft-radius landing pads
- Design non-sequential winch system to release payloads based on target requirements



HOW:

- Leveraged **electromagnetic clutches** to streamline deployment to single drive shaft
- Taught students **FEA** analysis in order to add light weighting features without compromising structural integrity
- Performed multiple prototype tests to optimize PID algorithm for drop speed

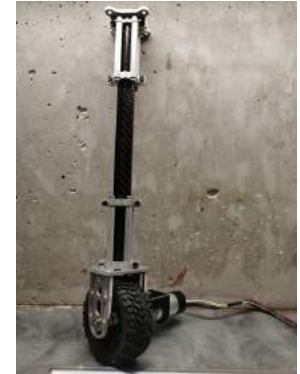
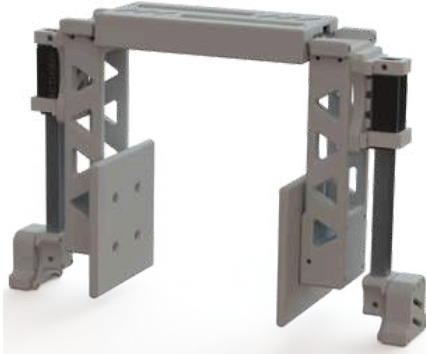
RESULTS:

- Successfully demonstrated EM clutch mechanism by independently dropping 5 water bottles non-sequentially
- Clutch control achieved minimum impact speed of $< 2\text{m/s}$ meeting competition requirements for a safe drop

DRONE PACKAGE RETRIEVAL SYSTEM - UBC Uncrewed Aircraft Systems (2022)

WHAT:

- Create adaptable clamping mechanism to retrieve and place packages up to 20x20x20cm and 2kg
- Aid in the design and manufacture of landing gear utilizing mecanum drive



HOW:

- Used **OnShape** to design lightweight claw mechanism using guide rails and **linear actuators**
- Iteratively optimized and manufactured landing gear from **waterjet cut** aluminum and carbon-fibre tubing
- Performed **FEA** on structural components to verify safety in operation

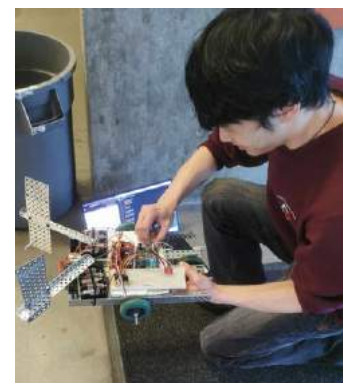
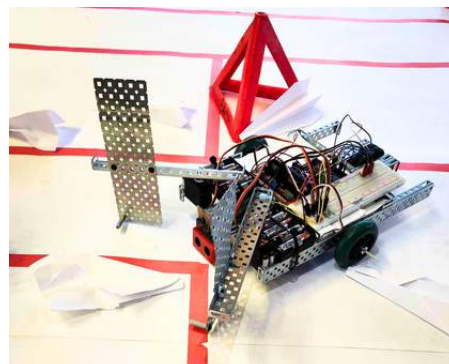
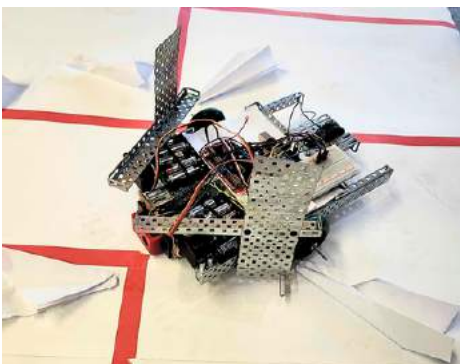
RESULTS:

- Adaptable claw was capable of retrieving 2kg packages while weighing 0.3kg
- Mecanum drive landing gear added freedom in ground mobility, allowing for leeway when landing near payload
- Modified claw mechanism successfully used to deliver football for the UBC 2023 homecoming game

AUTOMATED PAPER AIRPLANE COLLECTOR - Design Competition (2024)

WHAT:

- Robotics competition in a team of 4, given 8 hours to devise method to automatically locate, retrieve, and move paper airplanes to 'hangars'
- Provide technical presentation and full prototype demonstration to panel of judges and peers



HOW:

- Designed servo-actuated robot arm from limited materials (VEX Robotics Kits)
- Used Arduino to control servo drivers, motor control, and distance sensors
- Implemented rudimentary pathing to sweep the 'runway', and deliver airplanes to hangars

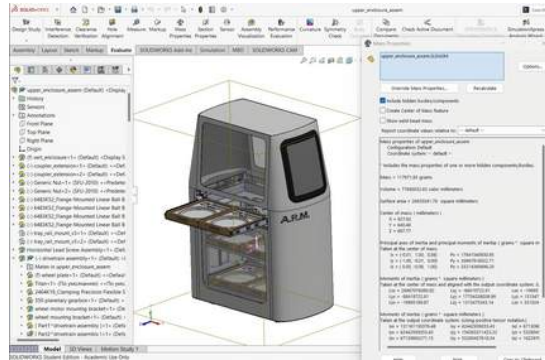
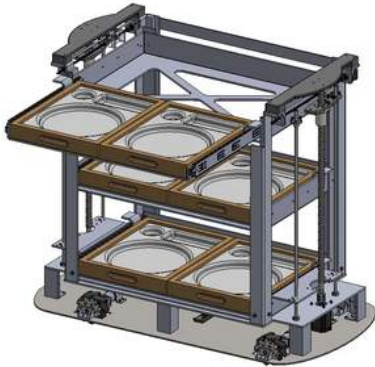
RESULTS:

- Autonomous retrieval robot successfully detected, collected, and returned airplanes to hangars with 90% accuracy
- Placed **1st of 5** teams for technical presentation and robot demonstration

RESTAURANT ROBOT DESIGN - Design Competition (2023)

WHAT:

- CAD competition in a team of 4, given 24 hours to develop means to improve automated restaurant food services
- Provide CAD models, technical presentation, and product pitch to panel of judges and peers



HOW:

- Exercised design process through 3 rounds of **concept sketches** of a restaurant robot
- Designed tray storage and dispensing mechanism in **SolidWorks** utilizing **sheet metal** features and **surface modelling**
- Provided detailed layout of subsystems through **2D technical drawings** and **BoM**

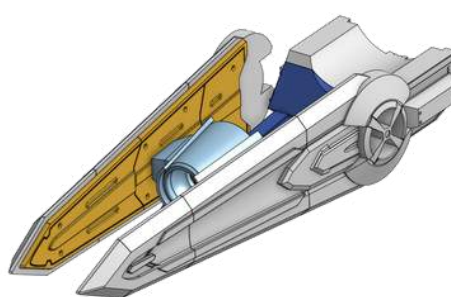
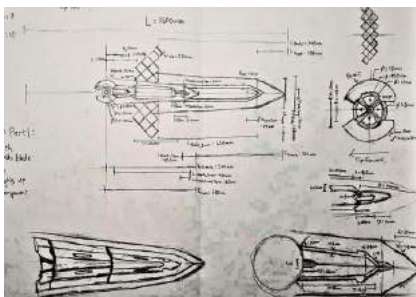
RESULTS:

- Drafted a fleshed-out **detailed design** within the allotted time including component speccking and **renders**
- Developed key skills in surface modelling, rendering components in **SolidWorks** and working under tight time constraints

COSPLAY DESIGN - Personal Project (Ongoing)

WHAT:

- Designing scale model functional cosplay pieces for 3D-printing with realistic moving components and lighting fixtures
- A passion project to practice compact component integration and Design for Manufacturing and Assembly (DFMA)



HOW:

- Create scaled **hand drawings** with minimal guidance by referencing promotional art and in-game assets
- Design components for **3D-printing** in **OnShape**, accounting for **DFMA**
- Exercise creativity while considering spatial constraints for integrating hardware components in **Arduino**

